

Azure Data Factory Dynamic Pipeline for Multi-Delimiter File Loading

Objective:

To build an automated Azure Data Factory (ADF) pipeline that dynamically loads multiple CSV files with different delimiters from Azure Data Lake Storage Gen2 (ADLS2) into Azure SQL Database tables, using metadata-driven configuration stored in a control table.

Steps and Setup:

1. Azure SQL Database Table Setup

- Created a control table named **File_Loading_List** in Azure SQL Database.
 - The table schema includes three columns:
 - **File_Name** (nvarchar) – the name of the source file.
 - **File_Delimiter** (nvarchar) – the delimiter character used in the respective CSV file.
 - **Target_Table** (nvarchar) – the target table in the database where the data should be loaded.
 - Inserted 3 rows into this table, each representing one CSV file with different delimiters (e.g., comma ,, pipe |, and semicolon ;).
 - Each CSV file had at least 6 fields and approximately 500+ rows of sample data sourced from public datasets.
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2. Data Preparation and Upload

- Downloaded 3 different CSV files from the internet, each using a distinct delimiter.
 - Uploaded these CSV files to an **Azure Data Lake Storage Gen2 (ADLS2)** container for use as source files in the pipeline.
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3. Azure Data Factory Pipeline Configuration

- Created a new ADF pipeline with dynamic processing capabilities as follows:

A. Lookup Activity

- Configured a **Lookup** activity to query the **File_Loading_List** table in Azure SQL Database.
- This lookup retrieves all rows (metadata) that define the file names, delimiters, and target tables.

B. ForEach Activity

- Used the output of the Lookup activity as an input array for a **ForEach** activity.
- This iterates through each row (file metadata) dynamically.

C. Copy Activity (Inside ForEach)

- Inside the ForEach loop, added a **Copy Data** activity.
 - Source: Configured to read the CSV file from ADLS2. The file path and delimiter are parameterized using the current ForEach item.
 - Sink: Writes data into the specified **Target_Table** in Azure SQL Database.
 - The delimiter for reading the CSV file is dynamically set based on the value from the **File_Delimiter** column.
 - Ensured the mapping between source columns and target table columns is appropriate.
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4. Supporting Configuration

- Set up linked services in ADF for:
 - Azure SQL Database connection
 - Azure Data Lake Storage Gen2 connection
 - Created datasets parameterized by file path and delimiter to allow dynamic loading.
 - Implemented pipeline parameters and expressions to handle dynamic file paths and delimiters.
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5. Validation

- Executed the pipeline and verified:
 - Successful pipeline runs in ADF monitoring.
 - Data is correctly loaded into the target tables in Azure SQL Database.
 - Data integrity checked by sampling rows and comparing with original source files.
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Deliverables / Evidence (Screenshots Provided):

- **File_Loading_List Table Screenshot:** Showing the control metadata with file names, delimiters, and target tables.
- **Uploaded Files on ADLS2 Screenshot:** Confirming the three CSV files with different delimiters are stored correctly.

- **ADF Pipeline Run Success Screenshot:** Showing the pipeline execution status as successful.
- **Lookup Activity Configuration Screenshot:** Displaying SQL query and settings to retrieve file metadata.
- **Copy Activity Configuration Screenshot:** Showing dynamic source settings including file path and delimiter and sink mapping.
- **Output Screenshot of Target Table:** Confirming data has been loaded successfully.

1. 3 files with different delimiters*

The screenshot shows the Microsoft Azure portal interface for a storage account named 'containersoumya'. The 'Containers' section is active, displaying a list of blobs. The authentication method is 'Access key' and the location is 'containersoumya'. The search results show three blobs:

Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
apple_quality.csv	3/2/2024, 11:15:41 PM	Hot (Inferred)		Block blob	280.4 KiB	Available
Europe Sales Records.csv	3/3/2024, 12:44:17 PM	Hot (Inferred)		Block blob	125.62 KiB	Available
social_media.csv	3/2/2024, 7:10:03 PM	Hot (Inferred)		Block blob	86.03 KiB	Available

2. Screenshot of table File_Loading_List

The screenshot shows the Azure Data Studio interface with a SQL query executed. The query is as follows:

```

1 select count(*) from [dbo].[Europesales];
2
3 Insert into [dbo].[azuresoumya] values ('apple.csv',' ','stg_apple');
4 Insert into [dbo].[azuresoumya] values ('sales.csv',' ','stg_sales');
5 Insert into [dbo].[azuresoumya] values ('social.csv',' ','stg_social');
6
7 Select * from [dbo].[azuresoumya];
8
9 delete from [dbo].[azuresoumya];

```

The results are displayed in a table named 'File_Loading_List' with the following columns: File_Name, File_Delimiter, and Target_Table.

File_Name	File_Delimiter	Target_Table
apple.csv	,	stg_apple
sales.csv	,	stg_sales
social.csv	,	stg_social

3. Screenshot of 3 files loaded to ADLS2

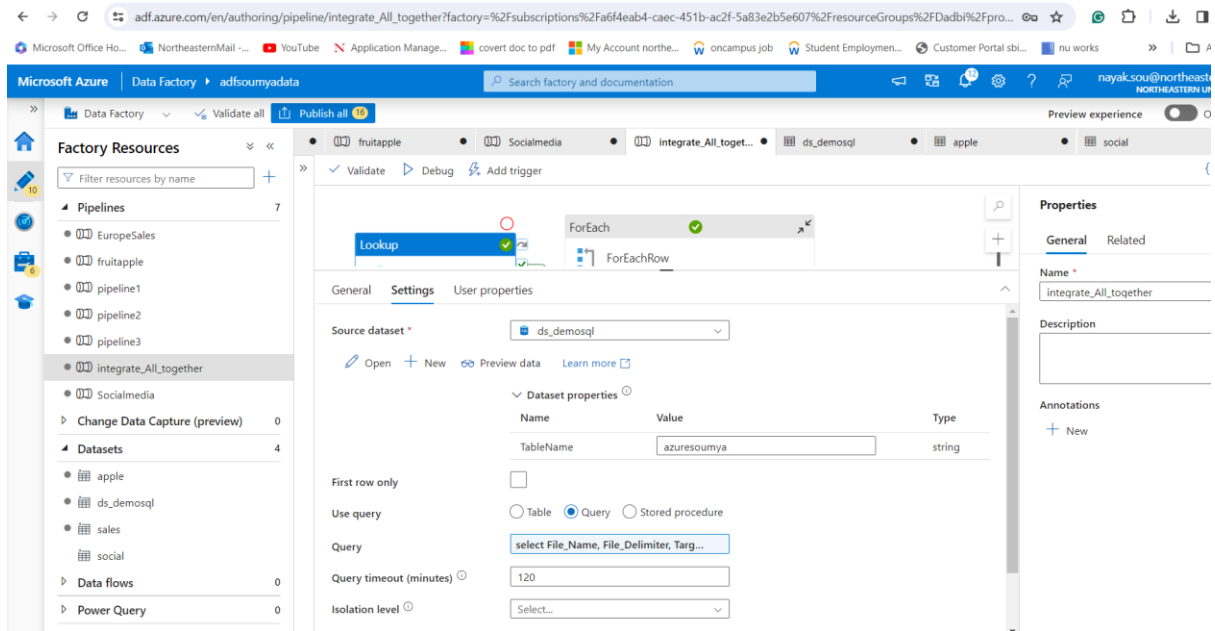
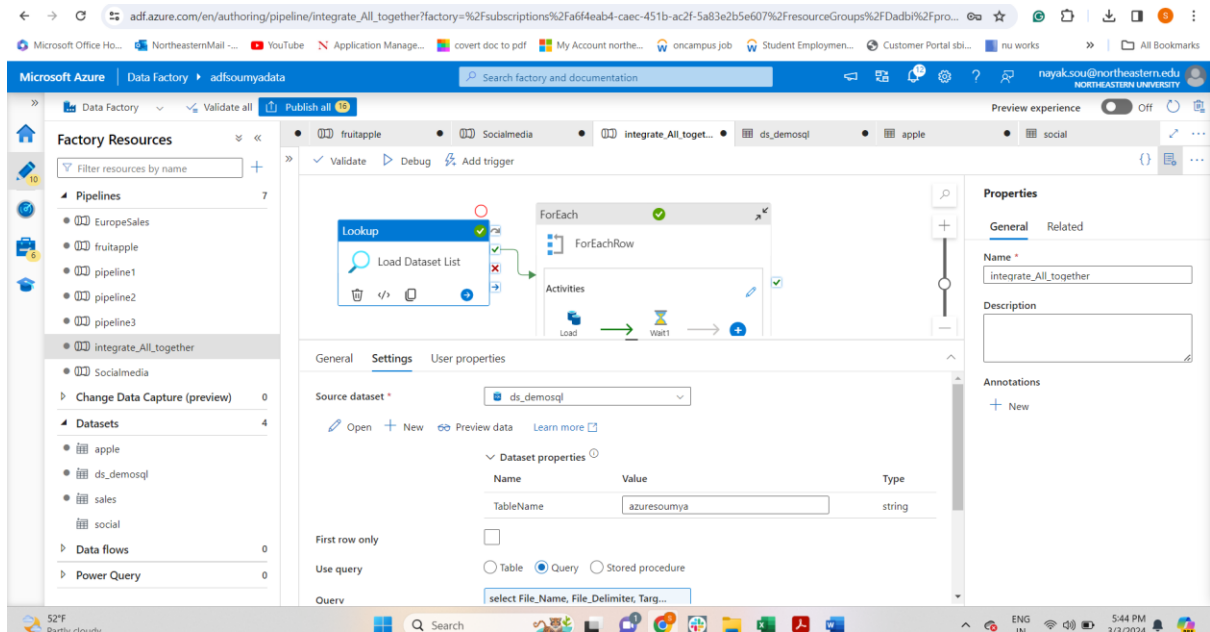
<a>↑ Upload <a>+ Add Directory <a>↻ Refresh <a>↶ Rename <a>🗑️ Delete <a>↔️ Change tier <a>🔒 Acquire lease <a>🔓 Break lease <a>🗨️ Give feedback							Summary
Authentication method: Access key (Switch to Microsoft Entra user account) Location: containersoumya							
Search blobs by prefix (case-sensitive) ☐ Show deleted objects							
Name	Modified	Access tier	Archive status	Blob type	Size	Lease state	
☐ apple.csv	3/2/2024, 11:15:41 PM	Hot (Inferred)		Block blob	280.4 KiB	Available	...
☐ sales.csv	3/2/2024, 7:10:03 PM	Hot (Inferred)		Block blob	86.03 KiB	Available	...
☐ social.csv	3/3/2024, 5:22:56 PM	Hot (Inferred)		Block blob	86.03 KiB	Available	...

4. Screenshot of Adf Pipeline successfull run

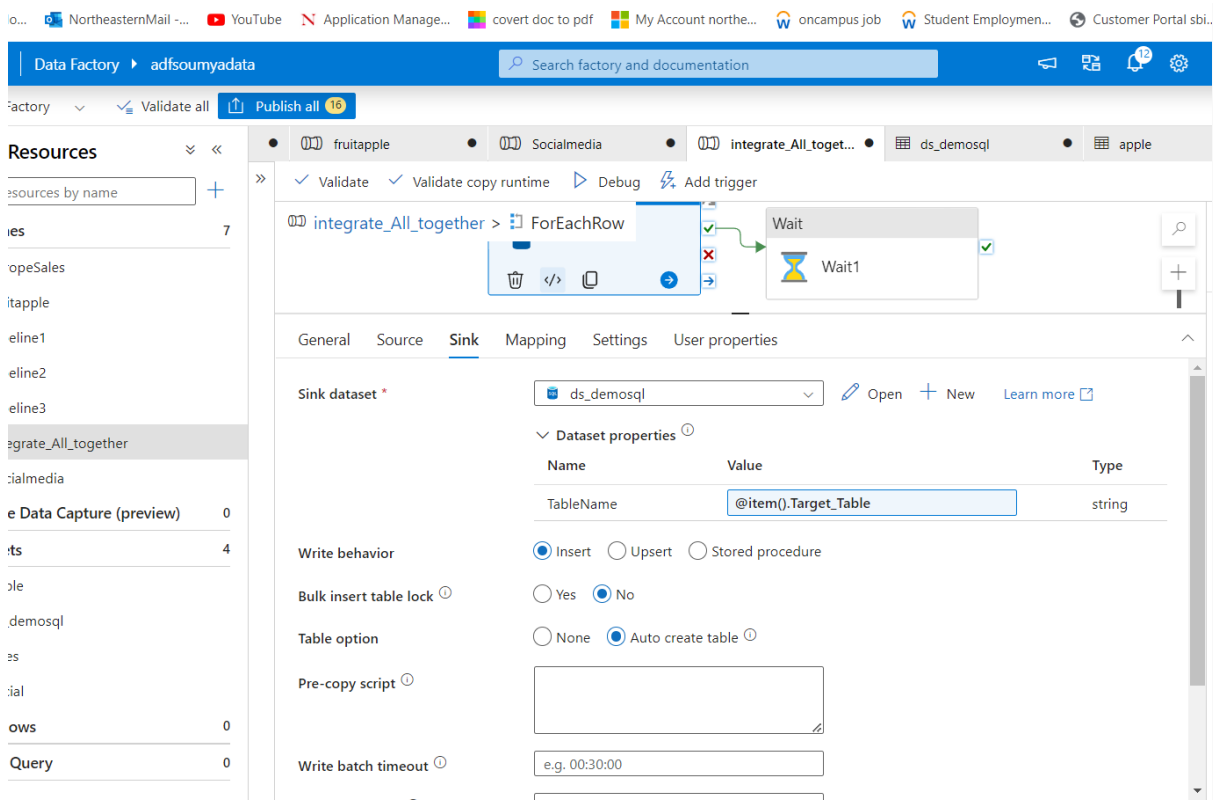
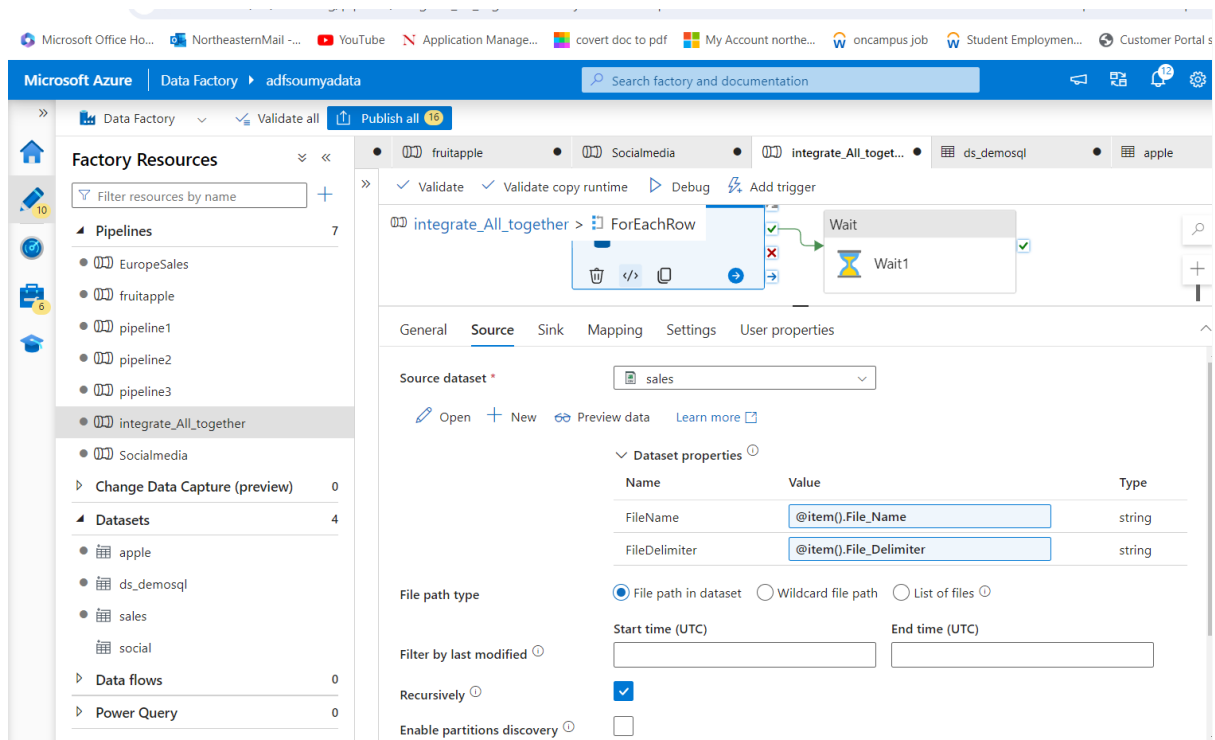
The screenshot displays the Microsoft Azure Data Factory interface. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'EuropeSales', 'fruitapple', 'pipeline1', 'pipeline2', 'pipeline3', and 'integrate_All_together' (selected). The main canvas shows the 'integrate_All_together' pipeline diagram, which includes a 'Lookup' activity connected to 'Load Dataset List', followed by a 'ForEach' loop containing a 'ForEachRow' loop with 'Load Dataset' and 'Wait1' activities. The right sidebar shows the 'Properties' pane for the selected pipeline. Below the canvas, the 'Output' tab is active, displaying a table of activity results.

Activity name	Activity status	Activity type	Run start	Duration	Ir
Wait1	Succeeded	Wait	3/3/2024, 5:39:00 PM	7s	
Load Dataset	Succeeded	Copy data	3/3/2024, 5:38:45 PM	14s	A
Wait1	Succeeded	Wait	3/3/2024, 5:38:38 PM	7s	
Load Dataset	Succeeded	Copy data	3/3/2024, 5:38:24 PM	13s	A
Wait1	Succeeded	Wait	3/3/2024, 5:38:17 PM	7s	
Load Dataset	Succeeded	Copy data	3/3/2024, 5:38:02 PM	14s	A

5. Screenshot of configuration of Lookup Activity



6. Screenshot of configuration of Copy Activity



7. Output Screenshot of Target table

Results Messages

Search to filter items...

Region	Country	ItemType	SalesChannel	OrderPriority
Europe	Czech Republic	Beverages	Offline	C
Europe	Bosnia and Herzegovina	Clothes	Online	M
Europe	Austria	Cereal	Offline	C

Query succeeded 10s

Query 1 × Query 2 × Query 3 × Query 4 ×

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT TOP (1000) * FROM [dbo].[SalesData]
```

Results Messages

Search to filter items...

Region	Country	ItemType	SalesChannel	OrderPriority
Europe	Czech Republic	Beverages	Offline	C
Europe	Bosnia and Herzegovina	Clothes	Online	M
Europe	Austria	Cereal	Offline	C

Azure Search resources, services, and docs (G+)

souserver/soumdb)

(souser/soumdb) | Query editor (preview)

Login New Query Open query Feedback Getting started

Tables

- AppleData.AppleData
- AppleData.SalesData
- dbo.AppleData
- dbo.azureoumya
- dbo.Europesales
- dbo.fruitapple
- dbo.fruitapple
- dbo.SalesData
- dbo.SocialMedia
- dbo.stg_apple
- dbo.stg_sales
- dbo.stg_social
- dbo.UserData
- dbo.UserData.UserData
- SalesData.AppleData
- SalesData.SalesData
- SalesData_v2.SalesData
- UserData.AppleData
- UserData.UserData

Query 1 × Query 2 × Query 3 ×

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT TOP (1000) * FROM [dbo].[stg_social]
```

Results Messages

Search to filter items...

age	gender	time_spent	platform	interests
56	male	3	Instagram	Sports
46	female	2	Facebook	Travel
32	male	8	Instagram	Sports

Microsoft Office Ho...NortheasternMail ~...YouTubeApplication Manage...covert doc to pdfMy Account northe...oncampus jobStudent Employmen...Customer Portal sbi...nu worksAll Bookmarks

Microsoft AzureSearch resources, services, and docs (G+)

nayak.sou@northeaster...NORTHEASTERN UNIVERSITY

Home > sounddb (souserver/sounddb)

sounddb (souserver/sounddb) | Query editor (preview) ☆ ...

SQL database

Search

LoginNew QueryOpen queryFeedbackGetting started

OverviewActivity logTagsDiagnose and solve problemsQuery editor (preview)SettingsCompute + storageConnection stringsPropertiesLocksData managementReplicasSync to other databasesIntegrationsAzure Synapse Link

Tables

AppleData.AppleData...AppleData.SalesData...dbo.AppleData...dbo.azureoumya...dbo.Europesales...dbo.fruitapple...dbo.fruitapple...dbo.SalesData...dbo.SocialMedia...**dbo.stg_apple**...dbo.stg_sales...dbo.stg_social...dbo.UserData...dbo.UserData.AppleData...SalesData.AppleData...SalesData.SalesData...SalesData_v2.SalesData...UserData.AppleData...UserData.UserData...

Query 1 ×Query 2 ×Query 3 ×Query 4 ×

RunCancel querySave queryExport data asShow only Editor

1 SELECT TOP (1000) * FROM [dbo].[stg_apple]

ResultsMessages

Search to filter items...

A_id	Quality	Acidity	Sweetness	Crunchiness	Juiciness	Rip
0	good	-0.491590483	5.346329613	-1.012008712	1.844900361	0.3
1	good	-0.722809367	3.664058758	1.588232309	0.853285795	0.8
2	bad	2.621636473	-1.738429162	-0.342615928	2.838635512	-0.1
3	good	2.308333343	4.334833843	0.889274746	2.638070484	0.1